

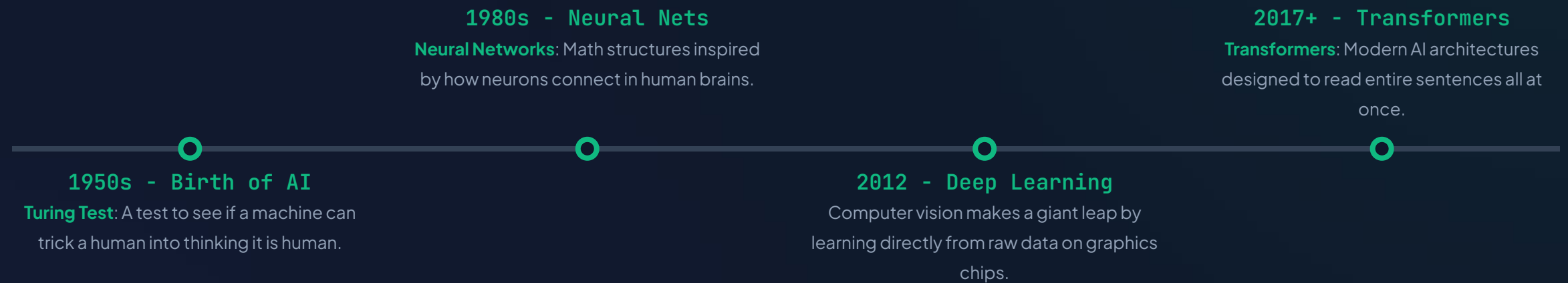
An Introduction to AI

A Friendly Guide to Models, Core Layers, and AI History

Part 1: The Evolution

How computer scientists went from building basic mathematical rules to creating smart programs that learn by themselves.

| Historical Foundations



DISCUSSION POINT

If you were playing a text-chat game, what is one question you would ask to figure out if you're talking to a real human or an AI?

| Symbolic vs Connectionist

Symbolic AI: "Rule-Based"

Think of this like an exact cooking recipe or math rulebook. Humans hand-write strict codes and logic trees (e.g., "If X is True, then do Y"). It works incredibly well for calculated math problems or chess, but breaks when it runs into messy, real-world data like blurry photos or slang.

Connectionist AI: "Learning-Based"

Instead of manual rules, we feed the program millions of examples and let it learn patterns on its own. Inspired by how biological brains practice and adapt. It figures out its own mathematical formulas to interpret real-world speech, images, and text.

QUICK CLASS DISCUSSION

Think about how you learn to play a video game. Do you read the entire instruction manual first (Symbolic), or do you just start playing and learn from your mistakes (Connectionist)?

| The 2012 Deep Learning Leap

2012

The ImageNet Breakthrough

Ditching Manual Coding

Before 2012, coders had to manually program features to detect objects (e.g., "find circles to detect eyes"). In 2012, a network called AlexNet won a major image-recognition contest by learning directly from raw images using **GPUs** (graphics processing units)—the fast computer chips originally created to render video games.

QUICK CLASS DISCUSSION

Can you think of an everyday task that is super easy to show someone how to do with examples, but really hard to write down a list of exact step-by-step rules for? (Hint: Think about emotions!)

| The 2017 Transformer Pivot

2017

Attention is All You Need

Ditching Recurrence for Speed

Older language systems read sentences step-by-step. If they read word 10, they started forgetting word 1. In 2017, scientists invented **Self-Attention**. This lets the computer read an entire sentence all at once and instantly connect matching words together (like connecting the pronoun "it" back to "the dog").

QUICK CLASS DISCUSSION

When you translate a sentence from Spanish to English, why is it better to read the whole sentence first instead of translating it word-by-word?

Part 2: Foundational Layers

Understanding how the AI universe is split into hardware, core brains, and user applications.

| The Three AI Stack Layers



1. Infrastructure

The Physical Engine: The raw computer chips (GPUs), massive cloud server rooms, and memory storage required to run calculations.



2. Model Layer

The Core Brain: The primary AI programs (like GPT-4, Llama, Gemini) that have learned language, logic, and patterns from internet data.



3. Application Layer

The User Tool: The software we build on top of the brain (like custom chat widgets, PDF readers, and tools) to solve actual user tasks.

QUICK CLASS DISCUSSION

If you wanted to build a cool new AI app for students, which of these three layers are you going to code yourself, and which layers are you going to rent from big tech companies?

| Model vs. Application Layer

The Model Space: The Generalist

Think of this as a college graduate who read the entire internet. It knows general grammar, facts, and code. However, it doesn't know your specific project instructions, private homework files, or university schedule yet.

The Application Space: The Specialist

This is where you build custom rules. You use tools like **RAG** (Retrieval-Augmented Generation) to feed the generalist model a specific textbook PDF, pointing it directly to the exact answers it needs to cite.

QUICK CLASS DISCUSSION

Instead of trying to train your own brand new AI model from scratch, why is it easier to just paste your class notes into ChatGPT and say "use this to quiz me"?

Part 3: Model Taxonomy

A simple breakdown of the different types of AI models and how they grow in size.

| Active Model Ecosystem



- Language Models / LLMs (50% - text chats & coding assistants)
- Computer Vision & Image Models (25% - image generation & detection)
- Multimodal (15% - reading audio, videos, and music)
- Traditional Tabular Models (10% - spreadsheets & predicting prices)

Different jobs require different styles of AI brains. While ChatGPT grabs the news, traditional tabular models are still run every day to crunch spreadsheet math and statistics.

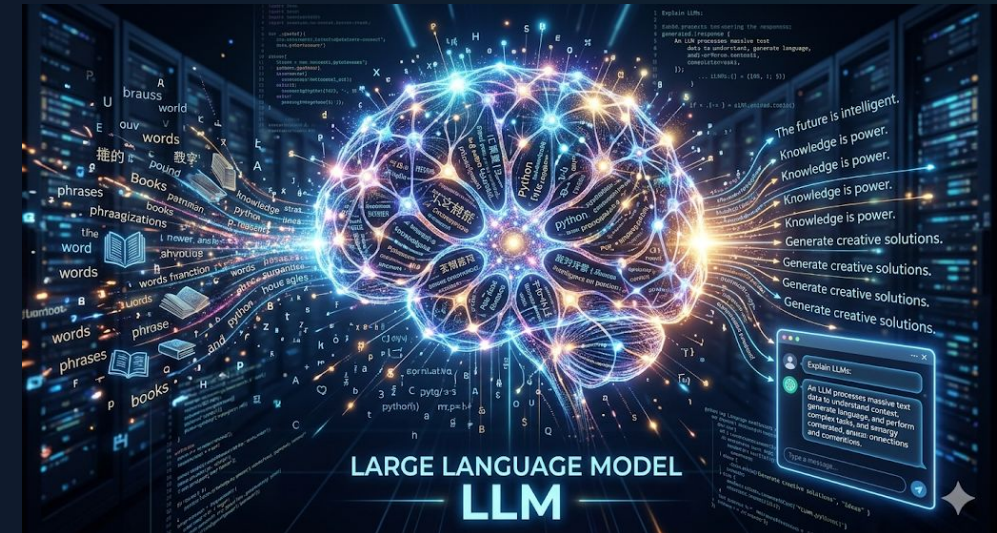
🗨️ QUICK CLASS DISCUSSION

We hear a lot about chat AI, but why might a self-driving car care way more about a Computer Vision model than a Language Model?

| Large Language Models (LLMs)

LLMs are like supercharged autocorrect. They look at a sentence and try to predict the next best word over and over again. They learn in three main stages:

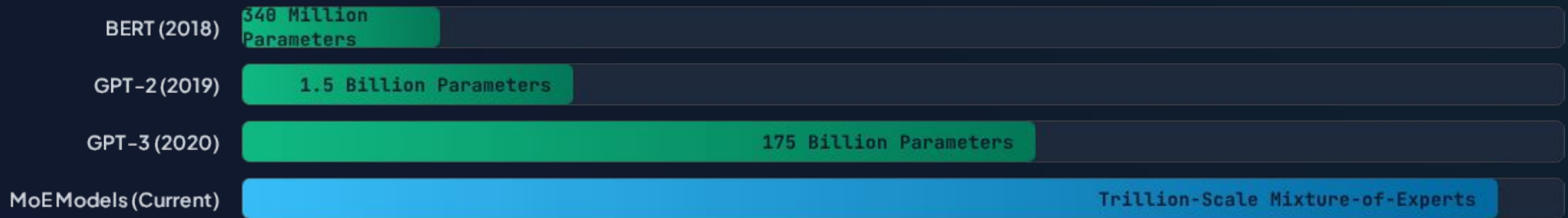
- **Pre-training:** Reading the internet to learn general vocabulary, syntax, and coding patterns.
- **Supervised Fine-Tuning (SFT):** Training the model with helpful Q&A scripts to act like an assistant rather than just completing sentences.
- **Alignment:** Installing boundaries so the AI refuses to answer harmful prompts.



🗣️ QUICK CLASS DISCUSSION

Have you ever had an AI helper refuse to answer your question, or respond with a super polite safety message? Which training stage is responsible for that?

| Scaling of Key Architectures



An AI's **Parameters** are like adjustable dials inside its digital brain. Adding more dials (parameters) allows the AI to learn vastly more complex math patterns and logic puzzles.

QUICK CLASS DISCUSSION

If more parameters make an AI brain smarter, why doesn't every company just make trillion-parameter models? (Hint: Think about energy bills, phone battery, and money!)

Questions & Discussion

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